Module 01 – Introduction to Cybersecurity (Revision Notes)

1. Digital Era

- Technology has become an essential part of everyday life.
- Most services are now connected through the internet.
- As digital usage increases, cyber threats also continue to grow.

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2. Cyber Attacks

- Cyber attacks target computers, networks, and digital information.
- They can lead to data theft, financial loss, and service disruption.
- Organizations use cybersecurity to defend against these attacks.

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3. What is Cybersecurity?

- Cybersecurity is the practice of protecting computers, networks, programs, and data.
- It prevents unauthorized access, attacks, and damage.
- It combines technologies, processes, and security practices.

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4. CIA Triad

The three core principles of cybersecurity are:

Confidentiality

- Protects information from unauthorized access.

Integrity

- Ensures that data remains accurate and unchanged.

Availability

- Ensures that authorized users can access data whenever required.

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5. Three Pillars of Cybersecurity

Cybersecurity is built on:

- People
- Processes
- Technology

These three pillars work together to maintain security.

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6. Incident

- A security incident is any event that compromises confidentiality, integrity, or availability.
- Incidents should be detected and handled quickly.

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7. Incident Response Team

- Responsible for identifying, analyzing, containing, and recovering from cyber incidents.
- Works to minimize damage and restore normal operations.

Course Learning Roadmap

8. Python & Linux Fundamentals

Topics include:

- Linux command line
- File permissions
- Shell scripting
- Python basics
- Web scraping
- Network scripting

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9. Foundations of Information Security

Topics include:

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- Cyber threats
- Ethical hacking
- CIA Triad
- Information gathering
- Reconnaissance
- Common cyber attacks

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10. Application Security & Penetration Testing

Topics include:

- SQL Injection
- Cross-Site Scripting (XSS)
- Metasploit
- DVWA
- BWAPP
- Mobile application security
- Vulnerability assessment

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11. Network Defense & Penetration Testing

Topics include:

- Network security
- Security protocols
- Firewalls
- Routers
- Switches
- Encryption
- Backup strategies
- Penetration testing
- Server hardening

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12. Governance, Risk & Compliance

Topics include:

- Security audits
- Risk management

- Compliance
- ISO
- ISMS

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13. Security Operations Center (SOC)

Topics include:

- SOC overview
- SIEM architecture
- Splunk
- Root Cause Analysis (RCA)
- Incident investigation
- Endpoint hardening

Kali Linux

14. What is Kali Linux?

- Debian-based Linux distribution.
- Designed for penetration testing and digital forensics.
- Maintained by Offensive Security.
- Includes numerous cybersecurity tools.

Cybersecurity Career Profiles

15. Computer Forensic Analyst

Role

- Investigates digital evidence.

- Recovers deleted or damaged data.
- Supports legal and corporate investigations.

Key Responsibilities

- Evidence collection
- Data recovery
- Digital analysis
- Documentation
- Report preparation
- Legal testimony

Technical Skills

- Windows, Linux, macOS
- NTFS, FAT32, EXT4
- EnCase
- FTK
- Autopsy
- Sleuth Kit
- Networking fundamentals

Industry Applications

- Law enforcement
- Corporate security
- Consulting firms

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16. SOC Analyst

Role

- Monitors security alerts.
- Detects cyber threats.
- Responds to security incidents.

Key Responsibilities

- Monitor security alerts.
- Perform incident detection.
- Respond to security events.
- Investigate suspicious activities.

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Career Path

L1

- Alert monitoring
- Initial investigation

L2

- Incident investigation
- Incident response

L3

- Threat intelligence
- Incident management
- Team leadership

Tools

- Splunk
- IBM QRadar
- IDS
- IPS
- Snort
- Firewalls
- Antivirus
- CrowdStrike (EDR)

Technical Skills

- Networking
- Windows
- Linux
- macOS
- SIEM tools

Key Takeaways

- Learned the fundamentals of cybersecurity.
- Understood the CIA Triad and the importance of protecting digital assets.
- Explored the overall course roadmap.
- Learned the purpose of Kali Linux.

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- Gained an introduction to cybersecurity career roles such as Computer Forensic Analyst and SOC Analyst.
- Became familiar with commonly used cybersecurity tools and technologies.